From Indifference to Dependence: Exploring User Engagement with AI Emotional Support Chatbots

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ABSTRACT

As digital emotional support has become more significant in society, artificial companionship has become increasingly popular. For some, bonds become so deep that they can lead to overdependence on their AI companions.

To explore the effects of mental well-being on human-robot bonding and dependence, we created an emotional support chatbot that includes a default personality as well as customizable personalities and analyzed human-robot attachment in participants of various mental/emotional states.

1 INTRODUCTION

As technological advancements continue to improve our daily lives, artificial companionship has become increasingly common. This has made its way into mental health support, benefitting many and promoting mental health awareness. Some, however, can become overly dependent on their AI chatbots.

Our study analyzes the effects of mental well-being on human-robot bonding and dependence. To do this, our team built an AI chatbot with default and customizable personalities and recruited participants for our study on this phenomena.

Over a period of one week, participants engaged with the default and customizable chatbot personalities at least once each, then were free to engage whenever desired. For the initial mandatory uses, one group interacted with the default personality first before custom, and the other group started with custom then moved to default. This measured whether a customizable personality would effect the depth of the bond between user and chatbot.

Before engaging with the bot, participants completed an initial survey inquiring about their mental and emotional state over the last two weeks. Once the study period ended, users completed a closing survey inquiring about their emotional attachment and feelings towards the chatbot.

Our study analyzes the research question:

What are the effects of mental well-being and mood on human-robot emotional dependence when engaging in artificial companionship?

Our study will test two hypotheses:

1. Human-robot emotional dependence will be higher when users' mental well-being is poor than when users' mental well-being is good when engaging in artificial companionship.

2. Human-robot emotional dependence will be higher when users can customize the AI chatbot than when users cannot customize the AI chatbot when engaging in artificial companionship.

2 RELATED WORK

2.1 Stigma Against Mental Health Services

Digital mental health services often face stigma that these services are for those who are mentally weak or have an "illness".

In many Eastern cultures, mental health issues are commonly stigmatized. Many peoples' mental well being included social pressure such as securing a good salaried job compared to peers and meeting the expectations of others. Due to this emphasis on others' opinions as influencing their own mental health, people are less likely to seek out mental health tools due to social stigma and the belief that doing so symbolizes weakness. To combat this, mental health apps should research cultural contexts and design these tools in a way that users don't feel it is for treatment of an "illness" [9].

In some tools, underperformance was cited as the top complaint from customers. Users stated that the robots underperformed in their advertised AI capabilities, causing a bad impression of human robot experience in the consumer space. This drives people away from seeking out AI support in the emotional or mental health category, due to the belief that it will not perform adequately [7].

To overcome stigma and increase user comfort with these services, digital tools can increase accessibility and personalization, leading to people being more likely to reach out for help and benefit from support.

Many neurodiverse people find it daunting to open up in an in-person therapy session as well as digital tools with bias towards neurotypical people. Because some new LLMs are tailored to neurodivergent thought patterns, autistic individuals favored them over typical LLMs due to it being much less overwhelming. By overcoming the bias for neurotypical people, those with autism were much more likely to use these LLMs for emotional support [1].

For those in emotionally draining careers such as frontline workers in gender based violence, emotional burnout can happen more frequently. Digital mental health interventions can provide a less daunting feel than face-to-face interaction, and encourage more mental health treatment. Participants in a study for helping frontline workers in gender based violence with their unique job stressors held positive and hopeful beliefs about the use of digital tools to help cope with the sensitive nature of their work. Due to the tools being tailored to provide more privacy and accessibility, workers felt comfortable using them to help with job related emotional stress and burnout [2].

Convenience is a very important aspect in most things in life, including the customizability of mental health support. Digital mental health kits that are tailored to users' personal needs and schedule are making emotional support more accessible, encouraging more people to seek help when needed. Because of this, users are more likely to use these tools to improve their mental health. When customization and convenience are at play, people that are in need of support are more likely to reach out and get the help they need [3].

In some societies, such as South Korea, mental health issues are prevalent, but engagement with support services remains particularly low due to this stigma as well as accessibility. By introducing an augmented reality (AR) mediated service that maintains anonymity between client and therapist, users are more likely to seek mental health and emotional support. When utilizing this anonymity, users' anxiety and depression scores decreased over time and viewed AR mediated therapy in a positive light. Resources like these that overcome cultural stigma are likely to be viewed positively and result in more people seeking the help they need [5].

Another hurdle to overcome is the struggle for some to verbalize their mental health issues, potentially leading to the avoidance of support. Large language models (LLMs) have shown the ability to provide tailored guidance to each individual through methods such as dynamically generated questions to identify users' struggles and needs. Due to the flexibility and agency users have through these LLMs, experiences were cited as favorable, increasing the likelihood of more popularity in use [6].

With artificial entities becoming increasingly popular and widespread today, options to tailor your robot to fit your emotional needs have also emerged. Results in a study on this topic showed that over time, the scores for bonding variables were above midpoint. Users gave the robot verbal encouragement and some physical assurance such as head pats. The emotional bond created between human and robot resulted in enjoyment of interactions and desire for future interactions [8].

This customizability can also affect users' to think less badly of their struggles and the impact on them as a person. By reducing this stigma through personalization, nearly half of the participants in this study cited that their interaction with the chatbot made them think through their responses more. They felt that interaction with the chatbot was thought provoking and enjoyable. Due to the positive nature of the experience, users were more likely to use artificial emotional support and see it in a positive light [4].

By tailoring digital mental health services to consider social stigma surrounding mental health, people will be more likely to seek help and improve their mental and emotional well-being.

2.2 Positive and Negative Effects of AI Companions on Mental Well-being

AI companions have been created to improve the mental well-being of many people at a high availability. Previous research has used AI and LLMs to support workplace roles that include a high amount of emotional labor and successfully aided in emotional regulation by demonstrating empathy [11]. Other research has shown that there are additional prosocial benefits deriving from feelings of helpfulness towards AI agents and participants became less lonely after conversing with the AI agents [15]. Many LLMs are created with the intention of benefiting users with supportive conversations, but some users have reported negative interactions with chatbots.

Minor adjustments in prompt engineering can influence these negative interactions with LLMs by unintentionally creating bias in the response given by the LLM by referencing certain brands, political parties, and nations more often which threatens user autonomy and trust [12]. Researchers have also described a taxonomy of negative interactions between humans and AI using conversations with an existing popular AI companion. The taxonomy includes six main categories including: harassment and violence, relational transgression, misinformation, verbal abuse and hate, substance abuse and self-harm, and privacy violations [10].

2.3 Personalization through Customization of AI Companions

Personalization of AI companions has become central in shaping how users form emotional connections. Emotional support chatbots that allow customization of memory, personality, and avatars encourage stronger bonds by fitting into the user's emotional and social needs. Participants in one study emphasized that the ability to configure a chatbot's demeanor, voice, or role was vital in making interactions meaningful and supportive [13]. This flexibility enables AI to serve diverse functions, such as confidants for vulnerable users or companions for casual reflection. However, while personalization enhances engagement, it also increases the risk of emotional dependence as users may begin to overly rely on the AI for comfort instead of human relationships.

Research on caregiving for robots provides additional insight into how human actions and emotional care towards AI affect bonding. Even when explicit self-reports did not indicate stronger ties, subtle behavioral cues such as hesitancy to replace a robot or increased effort in helping it showed evidence of attachment [8]. This suggests that people may unconsciously bond with AI through caregiving behaviors, even when they do not consciously recognize it. Personalization features that encourage "taking care" of AI companions might unintentionally amplify this attachment, raising concerns about fostering reliance that is not transparent to users.

The application of AI to mental health contexts further complicates this relationship. AI-driven tools that integrate causal knowledge graphs for depression stigma show how technology can uncover and reshape sensitive beliefs [4]. When combined with customizable companions, these systems risk entrenching dependency; users might prefer to confide in a personalized AI rather than challenge stigma and reach out to human support systems. While AI can play a crucial role in reducing barriers, designers must incorporate safety measures such as prompts to seek offline support, transparency about AI's limitations, and strategies to prevent excessive reliance.

Older adults, who are often identified as key beneficiaries of conversational AI, highlight the double-edged nature of personalization. Reviews reveal that seniors value trustworthiness, clear boundaries, and control in how AI adapts and remembers [14]. While tailored AI can provide companionship and reduce loneliness, it must not compromise autonomy or obscure its artificial nature. For this population, personalization must be balanced with mechanisms that prevent over-anthropomorphization. Together, these studies suggest that while customization deepens engagement and accessibility, it also requires careful design guidelines to ensure emotional bonds remain supportive rather than dependency inducing.

3 METHOD

3.1 Research Question and Hypotheses

Our study analyzes the research question:

What are the effects of well-being and mood on human-robot emotional dependence when engaging in artificial companionship?

Our team tested two hypotheses:

1. Human-robot emotional dependence will be higher when users' mental well-being is poor than when users' mental well-being is good when engaging in artificial companionship.

2. Human-robot emotional dependence will be higher when users can customize the AI chatbot than when users cannot customize the AI chatbot when engaging in artificial companionship.

We have built an emotional chatbot, 'MindMate', that specializes in providing users with emotional support and mental health advice for the purpose of improving mental and emotional well-being.

3.2 Chatbot Design

MindMate has a default personality as well as a customizeable personality option where users can enter a personality of their choice.

We have designed a default personality as well as a customizable option to analyze the emotional dependence level of users when using a general, predetermined personality vs. a personality type of their choice that they feel more comfortable with.

Figure 3.2.1 shows the default personality interface, while Figure 3.2.2 shows the custom personality interface where users can enter a personality of their choice.

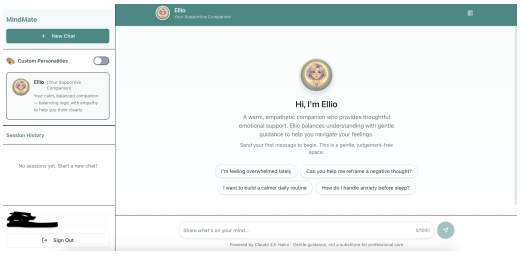

Figure 3.2.1: Default Personality Interface
This figure shows the default chatbot personality interface.

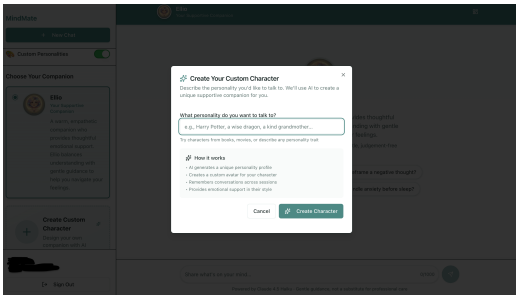

Figure 3.2.2: Custom Personality Selection
This figure shows the custom chatbot personality interface. Users are able to enter a personality of their choosing.

3.3 User Accessibility

MindMate is accessed through this link: https://emotional-support-chatbot-eight.vercel.app/

From there, users will be able to either use the default personality, as shown in Figure 3.3.1, or create a custom chatbot personality, as shown in Figure 3.3.2. Once the user has selected an AI companion, they can start interaction, as shown in Figure 3.3.3. Once finished with a session, users can either start a new chat session and/or clear their interaction history, as shown in Figure 3.3.4.

MindMate's task flow is as follows:

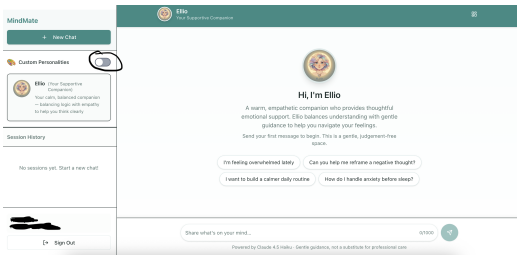

Figure 3.3.1: Toggle between Default and Custom Personality Interface
Once the user navigates to the chatbot interface, they can toggle the switch to access the customizable personality.

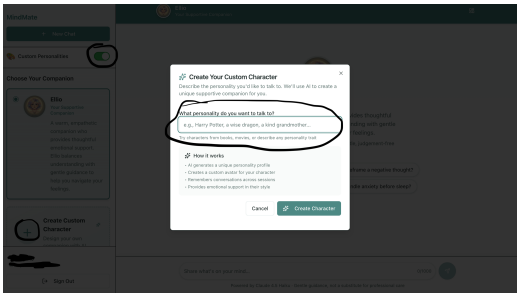

Figure 3.3.2: Create Desired Custom Personality
Once the user arrives at the custom personality interface, they can enter any personality they wish.

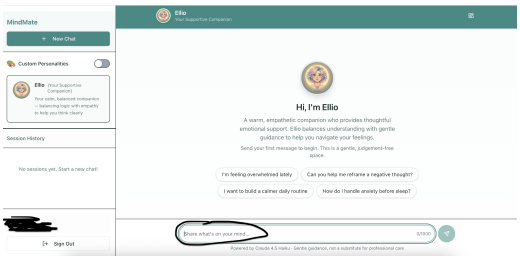

Figure 3.3.3: Begin Chatting
The user can now begin chatting by typing in the message bar.

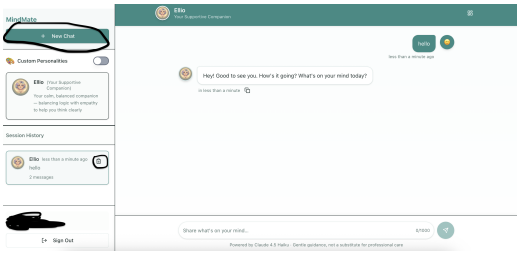

Figure 3.3.4: Start New Chat, Clear Current Chat History
To start a new chat, click on the '+ New Chat' button on the left sidebar.
To clear current chat history, click on the trash can symbol on the top right of the desired chat history entry under 'Session History'.

During the testing period, users will start with a mandatory one-time use of the default personality followed by a mandatory one-time use of the custom personality, or switched order of default and custom (one group for each). After this, users will have free reign on their usage frequency of any personality option (default or any customized) so their emotional dependence on the bot can be analyzed realistically.

4 USER STUDY

4.1 Research Question and Hypotheses

Our study explores the research question:

What are the effects of mental well being and mood on human-robot emotional dependence when engaging in artificial companionship?

We have two hypotheses to this question:

Hypothesis 1: Human-robot emotional dependence will be higher when users' mental well-being is poor than when users' mental well-being is good when engaging in artificial companionship.

Hypothesis 2: Human-robot emotional dependence will be higher when users can customize the AI chatbot than when users cannot customize the AI chatbot when engaging in artificial companionship.

4.2 Study Description and Design

Our chatbot has a default personality and a custom personality option. The custom option allows the user to create their own chatbot personality based on user input.

We are conducting a between-participants design, meaning our group will be split in two, with both groups being tested with different conditions.

For our study, this will be whether using the default personality first vs. using the custom personality first will affect users' emotional bond to the chatbot.

We have a total of twelve participants consisting of individuals aged 18-29. Six of our participants will be using the default personality first, then the custom personality. The other six of our participants will be using the custom personality first, then the default personality.

Starting Survey: Participants will first complete a starting survey that includes a consent form as well as the following question asking how they have been feeling in the last two weeks by rating each of the following statements:
I have felt cheerful and in good spirits.
I have felt calm and relaxed.
I have felt active and vigorous.
I woke up feeling fresh and rested.
My daily life has been filled with things that interest me.
Rating Scale:
0 - At no time
1 - Some of the time
2 - Less than half of the time
3 - More than half of the time
4 - Most of the time
5 - All of the time

4.3 Training Task

Users can watch the demo video we attached to the user instructions demonstrating how to use the chatbot. They will then be able to familiarize themselves with our interface as needed.

4.4 Participant-Chatbot Interaction

For the default first group, participants will send at least two messages to the default personality, then create a custom personality and send at least two messages to it.

For the custom first group, participants will create a custom personality and send at least two messages to it, then switch to the default personality and send at least two messages to it.

Then, for the rest of the study duration, all participants will use the chatbot whenever they want and with any of the personalities, including default and any old/new custom personalities.

Participants interacted with the chatbot from Friday (10/31/25) through Monday (11/3/25).

Concluding Survey: After this study period, users completed a concluding survey with the following questions:

1. How many different custom personalities did you create? Please respond with a number. The minimum is 1 since there was 1 mandatory chat session with a customized personality in the instructions.

2. Which chatbot personality do you prefer to chat with? Default or Customized.

3. What aspects of that chatbot personality made you prefer it?

4. How emotionally close do you feel to the DEFAULT PERSONALITY chatbot?
1 - Not at all close
2 - Slightly close
3 - Moderately close
4 - Very close
5 - Extremely close

5. How much do you agree or disagree with the following statements regarding the DEFAULT PERSONALITY chatbot?
I think about conversations with this chatbot when I'm not using it.
I use this chatbot when I need someone to talk to.
I rely on this chatbot for emotional support.
I sometimes prefer talking to this chatbot over talking to people I know.

Rating Scale:
Strongly disagree
Disagree
Neither agree nor disagree
Agree
Strongly agree

6. If the DEFAULT PERSONALITY chatbot was discontinued, how affected would you feel?
1 - Not at all affected
2 - Slightly affected
3 - Moderately affected
4 - Very affected
5 - Extremely affected

7. How emotionally close do you feel to the CUSTOM PERSONALITY chatbot?
1 - Not at all close
2 - Slightly close
3 - Moderately close
4 - Very close
5 - Extremely close

8. How much do you agree or disagree with the following statements regarding the CUSTOM PERSONALITY chatbot?
I think about conversations with this chatbot when I'm not using it.
I use this chatbot when I need someone to talk to.
I rely on this chatbot for emotional support.
I sometimes prefer talking to this chatbot over talking to people I know.
Rating Scale:
Strongly disagree
Disagree
Neither agree nor disagree
Agree
Strongly agree

9. If the CUSTOM PERSONALITY chatbot was discontinued, how affected would you feel?
1 - Not at all affected
2 - Slightly affected
3 - Moderately affected
4 - Very affected
5 - Extremely affected

4.5 Study Limitations

Due to our research group consisting of all master's students, this creates some limitations.

In particular, our participants are not very diverse, as they were all found through friends of our research group as well as classmates from our Human Computer Interactions course at The University of Texas at Dallas.

Also, our model charges money by number of messages, and because we are all students, we do not have the adequate funding to extensively test our hypotheses.

Another limitation is in regards to our participant split between default first and custom first. Some participants tried to interact with the chatbot right away before reading all the instructions. This may slightly skew our balancing initiative where we aim for an even split between default interaction first vs. custom interaction first.

Additionally, data could be skewed based on day to day factors and feelings outside of mental health reasons that affect the participants. Participants have varying amounts of interaction based on their own free use of the chatbot, so participants' schedules during the study period or external conditions may impact results.

A longer study may be needed to learn more about chatbot interactions over a longer period of time in order for a bond to form. Because of the short period of the study, results may be skewed to indifference because of lack of time spent with the chatbot.

4.6 Results

4.6.1 Descriptive Statistics. We have done multivariate statistical tests on our results, as we are analyzing how multiple variables interact with and affect each other. Specifically, how users' mental well-being and whether the chatbot's personality can be customized affect user-chatbot emotional bonding and dependence.

The following descriptive statistics were measured on the variables of users' initial emotional state vs. users' concluding emotional closeness to the chatbot: "Please rate how often you felt cheerful" vs. "How emotionally close to the chatbot".

Figure 4.6.1.1 Shows Pearsons's r, covariance, Spearman's rho, and Spearman's p-value. Figure 4.6.1.2 Shows a scatterplot of the correlation between the descriptive statistics variables the tests were conducted on.

Rate cheerful past 2 weeks	Emotional closeness chatbot	Pearson's r	Covariance	Spearman's rho	Spearman's p
Please rate how often you felt cheerful	How emotionally close to the chatbot	0.0978232	0.09090909	0.0554444	0.8641125

Figure 4.6.1.1: Table containing Descriptive Statistics

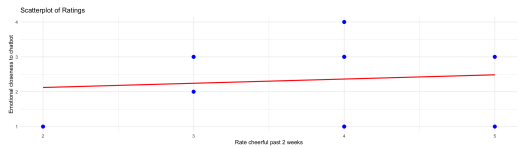

Figure 4.6.1.2: Scatterplot showing Correlation

As seen above, the Pearson's r was calculated to be 0.098. r(10) = 0.098, p = 0.098. This measures the extent to which the variables are proportional or linearly related to each other. Pearson's r has a range from [-1,1], with -1 indicating negative relation, 0 indicating no relation, and 1 indicating positive relation. As seen here, our value of 0.098 indicates a weak relation due to being close to 0, meaning that users' initial mental well-being is not proportional or linearly related to their concluding emotional bond with the chatbot.

Our covariance value came out to be 0.091. Covariance measures how two variables change together. If the covariance is positive, then when one variable increases, so does the other. If the covariance is negative, then when one variable increases, the other decreases. If the covariance is close to 0, then the variables are largely independent. Because our covariance is 0.091, this indicates a very

small positive association between the variables, contrary to what Pearson's r indicated.

Our Spearman's rho value was calculated to be 0.055. Spearman's rho (Spearman's rank correlation coefficient) measures whether ranks of variables move together and has a range of [-1,1], with -1 indicating negative correlation, 0 indicating no correlation, and 1 indicating positive correlation. Our value indicates a positive direction, but is extremely weak. This means that as emotional positivity increases, so does the bond with the chatbot, although very weakly.

Lastly, we analyze our Spearman's p-value, which was calculated to be 0.8641125. Spearman's p-value measures the statistical significance of correlation, with p < 0.05 being statistically significant and p >= 0.05 being insignificant. In our case, our Spearman's value indicates a very small positive correlation.

Overall, these four values indicated that there was a very small positive correlation between users' initial mental well-being and users' concluding emotional bond with the chatbot. Specifically, when users' initial mental well-being was better, their emotional bond with the chatbot increased.

This actually proves the opposite of our first hypothesis: 'Human-robot emotional dependence will be higher when users' mental well-being is poor than when users' mental well-being is good when engaging in artificial companionship'.

As for our overall research question: 'What are the effects of mental well-being and mood on human-robot emotional dependence when engaging in artificial companionship?', our descriptive statistics indicated that more positive mental well-being and mood resulted in more dependence on the chatbot.

4.6.2 Inferential Statistics. A repeated-measures ANOVA was performed to evaluate our second hypothesis: 'Human-robot emotional dependence will be higher when users can customize the AI chatbot than when users cannot customize the AI chatbot when engaging in artificial companionship'. Emotional dependence was operationalized through three Likert-scale outcomes ranging from 1 to 5: emotional closeness ("How emotionally close do you feel to this chatbot?"), engagement (capturing the extent of user interaction or contemplation of the chatbot when seeking companionship), and discontinuation impact ("If this chatbot were discontinued, how affected would you feel?"). Given that each participant evaluated both the default and customized personalities, personality was considered a within-subjects factor with two levels (Default vs. Custom), directly addressing the overarching research question regarding whether basic personalization features in AI companions substantively augment user-robot emotional bonds.

Descriptive statistics indicated that participants exhibited low to moderate dependence on both chatbot versions (refer to Figure 1). The mean ratings for emotional closeness was M = 2; 3 (SD approx 1.07) for the default personality and M = 2.17 (SD approx 0.94) for the custom personality on the 1–5 scale. The individual "spaghetti" plot in Figure 2 reveals considerable variability among participants: some felt more emotionally close to the custom chatbot (blue lines trending upwards), others favored the default (red lines trending downwards), and several showed no change. Nonetheless, these individual variations generally offset each other, resulting in comparable group means. Engagement ratings were modest overall, with a slight numerical bias favoring the custom personality (Default: M approx 2.25, Custom: M approx 2.67). Discontinuation impact scores were low across both conditions (Default: M approx 1.83, Custom: M approx 1.92), suggesting that most participants anticipated feeling only "not at all" or "slightly" affected if either version of the chatbot were to be discontinued.

Inferential analyses via repeated-measures ANOVAs indicated that none of these differences reached statistical significance. For emotional closeness, there was no main effect of personality: $F(1, 11) = 0.38$, p = .55, $\eta_p^2 \approx 0.03$, implying that participants did not perceive themselves as significantly closer to the customized chatbot compared to the default. Engagement showed a marginal increase for the custom personality, but this effect was small and not statistically significant: $F(1, 11) = 1.00$, p approx .34. Discontinuation impact differences were negligible: $F(1, 11) = 0.19$, p approx .67. Since personality had only two levels, the assumption of sphericity was trivially satisfied, and both visual inspection and normality checks of the difference scores supported the appropriateness of the repeated-measures ANOVA. Additionally, considering the three outcomes as a collective family of tests, applying a Bonferroni-corrected alpha level of .017 further confirmed that none of the effects of personality held much significance.

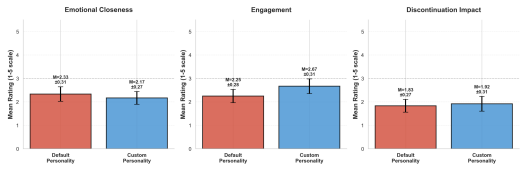

Figure 4.6.2.1: Mean ratings of emotional closeness, engagement, and anticipated discontinuation impact (1-5 scale) for the default (red) and custom (blue) chatbot personalities. Bars represent condition means, and error bars show ±1 SEM with numerical labels. Across all three outcomes, emotional dependence on the chatbot was generally low to moderate, with only a small, non-significant trend towards higher engagement in the custom personality condition.

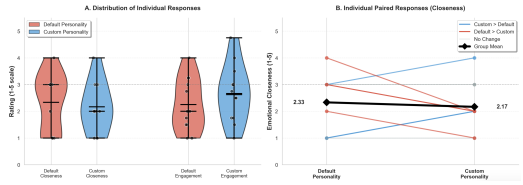

Figure 4.6.2.2: Individual differences in responses to the default vs. custom chatbot personalities. Panel A shows violin plots of the distributions for emotional closeness and engagement ratings (1-5 scale) under each personality, with overlaid individual data points and condition means. Panel B presents a slopegraph of paired emotional closeness ratings for each participant, with line colors indicating whether closeness increased for the custom personality (blue), decreased (red), or stayed the same (gray); the thick black line depicts the change in the group mean. Together, these panels highlight substantial variability in how personality customization affected perceived closeness and engagement.

Collectively, these findings do not substantiate Hypothesis 2 and provide a nuanced perspective on our research inquiry. At the group level, granting participants the ability to influence the chatbot's personality did not reliably enhance emotional closeness, daily engagement, or perceived impact of discontinuation. The individual variation observed in Figure 2 suggests that personalization might be beneficial for certain users by fostering increased connection, while others may remain unaffected or prefer the default configuration - a pattern that averages out in this limited sample. Consequently, personality customization alone within the short duration of our study proved insufficient in eliciting substantive or consistent emotional dependence on the AI companion. Future research involving larger samples and extended usage periods may be necessary to detect more subtle personalization effects or to identify which users are more likely to develop stronger bonds with tailored chatbot personalities.

We used a between-subjects design for balancing the order in which the participants used the chatbot personalities. Half of the participants used the default personality first and the custom personality second, while the other half of participants used the custom personality first and the default personality second. To determine if the ordering of chatbot personalities impacts the results, we performed unpaired t-tests on each of the dependent variables given from the concluding survey. As seen in the unpaired t-test results, none of the p-values were under 0.05. Thus, we can conclude that the ordering of personality use does not have a significant impact on the survey outcome. Balancing the ordering was necessary to ensure that the ordering of personality use did not significantly skew the data.

Unpaired t-test against ordering of personality used	t-value	d	p-value
Number of custom personalities created	0	10	1
Personality preference between default and custom	0	10	1
Closeness to default personality	-0.52	9.55	0.615
Think about default personality when not using chatbot	-0.45	9.86	0.6644
Use default personality when need someone to talk to	-0.73	9.42	0.4845
Rely on default personality for emotional support	-0.75	9.67	0.4738
Prefer talking to default personality over people they know	-1.14	9.06	0.2852
Affected if default personality was discontinued	-0.60	9.25	0.5644
Closeness to custom personality	-0.60	9.25	0.5644
Think about custom personality when not using chatbot	-0.23	7.03	0.8269
Use custom personality when need someone to talk to	-0.70	9.51	0.5033
Rely on custom personality for emotional support	-0.52	9.55	0.615
Prefer talking to custom personality over people they know	-0.87	9.51	0.4064
Affected if custom personality was discontinued	-0.79	9.59	0.4513

Figure 4.6.2.3: Unpaired T-Test

5 DISCUSSION

To reiterate, our study analyzes the research question:

What are the effects of mental well-being and mood on human-robot emotional dependence when engaging in artificial companionship?

Our team tested two hypotheses:

1. Human-robot emotional dependence will be higher when users' mental well-being is poor than when users' mental well-being is good when engaging in artificial companionship.

2. Human-robot emotional dependence will be higher when users can customize the AI chatbot than when users cannot customize the AI chatbot when engaging in artificial companionship.

As seen in our results and statistical analyses, no significant correlation was found.

In fact, our first hypothesis was disproven, with the opposite of what we predicted actually happening: users with more positive mental well-being had slightly higher dependence on the chatbot than those with comparatively poor mental well-being.

As for our second hypothesis, while the customizeable chatbot personality showed slightly higher preference than the default one, the bond and dependence felt from participants did not differ between the two.

While our first hypothesis showed a slight trend, due to the very weak statistical correlation, we draw the conclusion that mental well-being and mood have no significant effect on human-robot emotional dependence when engaging in artificial companionship.

6 CONCLUSION

In summary, this paper detailed a study on the effect of mental well-being and mood on human-robot emotional dependence when engaging in artificial companionship.

We found that while there was a very slight positive correlation between positive mental well-being and chatbot dependence (more positive mental well-being leading to more dependence), the statistical correlation was so small that it was deemed insignificant.

Therefore, we found that for our participants, mental well-being and mood had virtually no effect on human-robot bonding and dependence.

Due to limitations in participant size and time constraints, future work with more extensive studies would be more beneficial in exploring this topic.

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